Decentralized Graph Neural Network-Based Joint Beamforming in Multi-RIS-Aided Cell-Free Networks

Wen-Yuan Ting[1], Ronald Y. Chang[1], Feng-Tsun Chien[2], Tzu-Yuan Peng[1], and Pin-Han Lin[1]
[1]Research Center for Information Technology Innovation, Academia Sinica, Taipei, Taiwan
[2]Institute of Electronics, National Yang Ming Chiao Tung University, Hsinchu, Taiwan
Email: dingwenyuan22@gmail.com, rchang@citi.sinica.edu.tw, ftchien@nycu.edu.tw,
tzuyuanpeng0716@gmail.com, linhank0710@gmail.com

***Abstract*—Cell-free systems have emerged as a promising alternative to conventional cellular architectures. However, centralized implementations can incur significant overhead due to channel state information (CSI) exchange between access points (APs) and the central processing unit (CPU), as well as the distribution of beamforming solutions. In this work, we propose a decentralized graph neural network (GNN)-based framework for multiple reconfigurable intelligent surface (RIS)-assisted cell-free networks. Our approach distributes learning tasks across APs using only local CSI, where each AP executes a copy of the trained GNN to compute its beamforming vectors and RIS phase estimates, which are then aggregated at the CPU to determine the final RIS configuration. Simulations and analysis show that the proposed decentralized method achieves near-centralized sum-rate performance while reducing CSI acquisition and signaling overhead.**

***Index Terms*—Cell-free network, reconfigurable intelligent surface, decentralized beamforming, graph neural network.**

I. Introduction

User-centric cell-free networks have been proposed [1], [2] to enable ubiquitous connectivity across a wide service area. Unlike conventional cellular systems, users in cell-free networks are jointly served by multiple distributed access points (APs) connected to a central processing unit (CPU) via fronthaul links, without cell boundary restrictions. Such architectures have been shown to outperform traditional cellular systems in both spectral and energy efficiency [1].

Reconfigurable intelligent surface (RIS)-aided cell-free networks have been studied to further enhance network performance by exploiting the ability of RISs to engineer favorable propagation conditions, using both model-based [3]–[8] and learning-based approaches [9]–[12]. In [3], joint AP beamforming and discrete RIS phase shift design for weighted sum-rate maximization was studied by decoupling the problem into passive and active beamforming subproblems, solved via manifold optimization and fractional programming, respectively. In [4], joint AP and RIS beamforming for RIS-aided cell-free networks was designed to maximize the sum rate by decoupling the problem and solving it iteratively, and in [5], joint hybrid AP beamforming and RIS phase shift design for weighted sum-rate maximization was addressed using a block coordinate descent (BCD)-based algorithm. In [6], joint AP clustering and active-passive beamforming in RIS-aided cell-free networks was studied to maximize weighted sum rate while minimizing the number of clustered APs to reduce fronthaul overhead, using a BCD-based approach. In [7], joint RIS-user association and AP/RIS beamforming for weighted sum-rate maximization was studied to alleviate channel acquisition overhead, with the problem decoupled into association and beamforming subproblems and solved accordingly. In [8], joint AP active beamforming and RIS passive beamforming was studied for weighted sum-rate maximization in wideband RIS-aided cell-free networks, accounting for RIS frequency selectivity, and solved via alternating optimization.

For learning-based optimization of RIS-assisted cell-free systems, in [9], a sum-rate maximization problem was formulated for joint AP active and RIS passive beamforming under quality-of-service (QoS) constraints, and a centralized soft actor-critic (SAC)-based algorithm was proposed. In [10], joint AP beamforming and RIS phase shift design under hardware impairments and channel aging was addressed via deep reinforcement learning (DRL). In [11], AP selection, AP power control, and beyond-diagonal RIS design were jointly optimized using successive convex approximation and DRL. In [12], a heterogeneous graph neural network (GNN) architecture was proposed to optimize RIS phase shifts and user power control for improved uplink performance in RIS-assisted cell-free systems.

The aforementioned works focus on *centralized* joint optimization for RIS-aided cell-free systems, which incurs high signaling and computational overhead and limits scalability. To overcome this, *decentralized* frameworks for joint AP beamforming and RIS reflection coefficient optimization have been developed [13]–[15]. In [13], joint AP beamforming and RIS reflection optimization for sub-connected active RIS-assisted cell-free systems was addressed via a two-stage distributed

This work was supported in part by the National Science and Technology Council, Taiwan, under Grants NSTC 112-2221-E-001-004-MY3, NSTC 115-2221-E-001-008-MY3, and NSTC 115-2221-E-A49-075-MY2.

iterative algorithm to reduce CPU burden. In [14], a decentralized cooperative beamforming framework was developed by reformulating the centralized weighted sum-rate maximization problem as a consensus problem and solving it using an alternating direction method of multipliers (ADMM)-based algorithm. In [15], unsupervised deep neural networks (DNNs) were proposed for decentralized beamforming with limited AP–CPU communication overhead; however, RISs were not considered, which introduces additional complexity to the problem.

This work proposes a decentralized GNN-based framework for sum-rate maximization in multi-RIS-aided cell-free networks, motivated by the strong performance of GNNs in beamforming optimization for millimeter wave (mmWave) [16], [17] and RIS-assisted systems [18], [19], yet their limited exploration in decentralized joint beamforming for RIS-aided cell-free networks. In the proposed framework, each AP runs a copy of the trained GNN with locally available channel state information (CSI) to independently compute its beamforming vectors and RIS phase shift estimates, which are then aggregated at the CPU to obtain the final RIS configuration. Simulation results show that the proposed decentralized method closely matches the performance of its centralized counterpart, while reducing channel acquisition and signaling overhead.

Notation: $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$, and $(\cdot)^{\mathrm{H}}$ denote complex conjugate, transpose, and conjugate transpose operators, respectively. $\mathrm{vec}(\mathbf{X})$ vectorizes matrix $\mathbf{X}$, and $\mathrm{diag}(\mathbf{x})$ forms a diagonal matrix with the diagonal elements being the entries of $\mathbf{x}$. $\mathbb{E}[\cdot]$ denotes the expectation operator. $\mathrm{tr}(\mathbf{X})$ denotes the trace of matrix $\mathbf{X}$. $\|\cdot\|_2$ denotes the 2-norm. $\mathrm{Re}\{\cdot\}$ and $\mathrm{Im}\{\cdot\}$ represent the real and imaginary parts of a complex-value argument.

II. System Model and Problem Description

We consider a frequency division duplex (FDD) cell-free network serving multiple users with the assistance of multiple RISs. The system comprises L APs, R RISs, and K user equipments (UEs), where all APs are connected to a CPU, as shown in Fig. 1. Each AP is equipped with M antennas, each RIS consists of N passive elements, and each UE is equipped with a single antenna. Let $\mathcal{L} = \{1, 2, \dots, L\}$, $\mathcal{R} = \{1, 2, \dots, R\}$, and $\mathcal{K} = \{1, 2, \dots, K\}$ denote the sets of all the APs, RISs, and UEs, respectively. Moreover, let $\mathcal{L}_k$ be the set of APs serving the k-th UE, and $\mathcal{K}_l$ the set of UEs served by the l-th AP. We assume that each UE is served by at least one AP, while not all APs are required to provide service.

Let s_k be the symbol intended for the k-th UE from its serving APs, where $\mathbb{E}[|s_k|^2] = 1$ and $\mathbb{E}[s_k] = 0, \forall k$, and $\mathbb{E}[s_k s_i^*] = 0, \forall k \neq i$. Let $\mathbf{f}_{(l,k)} \in \mathbb{C}^{M\times 1}$ be the beamforming vector for the k-th UE at the l-th AP. The transmitted signal at the l-th AP can be expressed as $\mathbf{x}_l = \sum_{k\in\mathcal{K}_l} \mathbf{f}_{(l,k)} s_k$, with power constraint $\mathbb{E}[\|\mathbf{x}_l\|^2] \leq P_{\max,l}$.

Let $\boldsymbol{\theta}_r \triangleq \left[\theta_{(r,1)}, \dots, \theta_{(r,N)}\right]^{\mathrm{T}} \in \mathbb{C}^{N\times 1}$ be the phase shift vector at the r-th RIS, and let $\mathbf{G}_{(l,r)} \in \mathbb{C}^{N\times M}$, $\mathbf{h}_{\mathrm{R},(r,k)} \in \mathbb{C}^{1\times N}$, and $\mathbf{h}_{\mathrm{D},(l,k)} \in \mathbb{C}^{1\times M}$ denote the channels from the l-th AP to the r-th RIS, from the r-th RIS to the k-th UE,

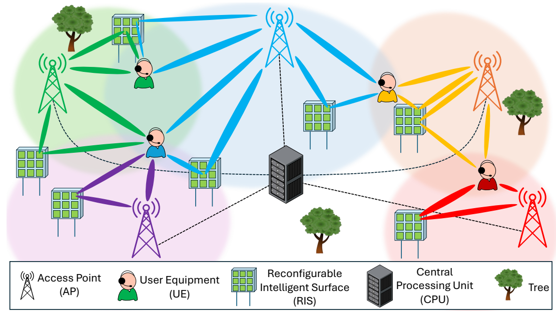

Fig. 1. A multi-RIS-aided multiuser cell-free network.

and from the l-th AP to the k-th UE, respectively. $\mathbf{G}_{(l,r)}$ is modeled as a Rician fading channel:

$$\begin{aligned}\mathbf{G}_{(l,r)} &= \bar{\eta}_{\mathrm{AR},(l,r)} d_{\mathrm{AR},(l,r)}^{-\bar{\gamma}_{\mathrm{AR},(l,r)}} \sqrt{\frac{\kappa_{\mathrm{AR},(l,r)}}{\kappa_{\mathrm{AR},(l,r)}+1}} \mathbf{a}_{\mathrm{R}}(\theta_{\mathrm{a},(l,r)}) \mathbf{a}_{\mathrm{A}}^{\mathrm{H}}(\theta_{\mathrm{d},(l,r)}) \\ &\quad + \bar{\bar{\eta}}_{\mathrm{AR},(l,r)} d_{\mathrm{AR},(l,r)}^{-\bar{\bar{\gamma}}_{\mathrm{AR},(l,r)}} \sqrt{\frac{1}{\kappa_{\mathrm{AR},(l,r)}+1}} \bar{\bar{\mathbf{G}}}_{(l,r)}, \end{aligned} \tag{1}$$

where $\mathbf{a}_{\mathrm{R}}(\theta_{\mathrm{a},(l,r)}) = [1, \dots, e^{j(N-1)\pi\sin(\theta_{\mathrm{a},(l,r)})}]^{\mathrm{T}} \in \mathbb{C}^N$ and $\mathbf{a}_{\mathrm{A}}(\theta_{\mathrm{d},(l,r)}) = [1, \dots, e^{j(M-1)\pi\sin(\theta_{\mathrm{d},(l,r)})}]^{\mathrm{T}} \in \mathbb{C}^M$ denote the RIS and AP steering vectors corresponding to the angle of arrival (AoA) $\theta_{\mathrm{a},(l,r)}$ and angle of departure (AoD) $\theta_{\mathrm{d},(l,r)}$, respectively; $\kappa_{\mathrm{AR},(l,r)}$ is the Rician factor; the non-line-of-sight (NLoS) component $\bar{\bar{\mathbf{G}}}_{(l,r)} \in \mathbb{C}^{N\times M}$ has i.i.d. entries distributed as $\mathcal{CN}(0,1)$; $\bar{\eta}_{\mathrm{AR},(l,r)}$ and $\bar{\bar{\eta}}_{\mathrm{AR},(l,r)}$ denote the large-scale fading gains of the line-of-sight (LoS) and NLoS components, respectively; $\bar{\gamma}_{\mathrm{AR},(l,r)}$ and $\bar{\bar{\gamma}}_{\mathrm{AR},(l,r)}$ are corresponding path-loss exponents; and $d_{\mathrm{AR},(l,r)}$ is the AP–RIS distance. Similarly, $\mathbf{h}_{\mathrm{R},(r,k)}$ is modeled as

$$\begin{aligned}\mathbf{h}_{\mathrm{R},(r,k)} &= \bar{\eta}_{\mathrm{RU},(r,k)} d_{\mathrm{RU},(r,k)}^{-\bar{\gamma}_{\mathrm{RU},(r,k)}} \sqrt{\frac{\kappa_{\mathrm{RU},(r,k)}}{\kappa_{\mathrm{RU},(r,k)}+1}} \mathbf{a}_{\mathrm{R}}^{\mathrm{H}}(\theta_{\mathrm{d},(r,k)}) \\ &\quad + \bar{\bar{\eta}}_{\mathrm{RU},(r,k)} d_{\mathrm{RU},(r,k)}^{-\bar{\bar{\gamma}}_{\mathrm{RU},(l,r)}} \sqrt{\frac{1}{\kappa_{\mathrm{RU},(r,k)}+1}} \bar{\bar{\mathbf{h}}}_{\mathrm{R},(r,k)}, \end{aligned} \tag{2}$$

where $\mathbf{a}_{\mathrm{R}}(\theta_{\mathrm{d},(r,k)}) = [1, \dots, e^{j(N-1)\pi\sin(\theta_{\mathrm{d},(l,r)})}]^{\mathrm{T}} \in \mathbb{C}^N$, $\bar{\bar{\mathbf{h}}}_{\mathrm{R},(r,k)} \in \mathbb{C}^{1\times N}$ has i.i.d. entries distributed as $\mathcal{CN}(0,1)$, $d_{\mathrm{RU},(r,k)}$ is the RIS–UE distance, and the remaining parameters are defined similarly to (1). The direct channel $\mathbf{h}_{\mathrm{D},(l,k)}$ is modeled as a Rayleigh fading channel:

$$\mathbf{h}_{\mathrm{D},(l,k)} = \bar{\bar{\eta}}_{\mathrm{D},(l,k)} d_{\mathrm{D},(l,k)}^{-\bar{\bar{\gamma}}_{\mathrm{D},(l,k)}} \bar{\bar{\mathbf{h}}}_{\mathrm{D},(l,k)}, \tag{3}$$

where $\bar{\bar{\mathbf{h}}}_{\mathrm{D},(l,k)}$ has i.i.d. entries distributed as $\mathcal{CN}(0,1)$, $\bar{\bar{\eta}}_{\mathrm{D},(l,k)}$ and $\bar{\bar{\gamma}}_{\mathrm{D},(l,k)}$ are the large-scale fading gain and path-loss exponent, respectively, and $d_{\mathrm{D},(l,k)}$ is the AP–UE distance.

The received signal at the k-th UE is given by

$$y_k = \sum_{l\in\mathcal{L}} \Big(\mathbf{h}_{\mathrm{D},(l,k)} + \sum_{r\in\mathcal{R}} \mathbf{h}_{(l,r,k)}\Big)\mathbf{x}_l + z_k, \tag{4}$$

where $\mathbf{h}_{(l,r,k)} \triangleq \mathbf{h}_{\mathrm{R},(r,k)}\mathrm{diag}(\boldsymbol{\theta}_r)\mathbf{G}_{(l,r)} \in \mathbb{C}^{1\times M}$ and $z_k \sim \mathcal{CN}(0,\sigma_k^2)$ is the additive white Gaussian noise (AWGN). The reflected channel $\mathbf{h}_{(l,r,k)}$ can be reformulated as $\mathbf{h}_{(l,r,k)} = \left(\mathbf{H}_{(l,r,k)}^{\mathrm{T}}\boldsymbol{\theta}_r\right)^{\mathrm{T}}$, where $\mathbf{H}_{(l,r,k)}^{\mathrm{T}} \triangleq \mathbf{G}_{(l,r)}^{\mathrm{T}}\mathrm{diag}(\mathbf{h}_{\mathrm{R},(r,k)}) \in \mathbb{C}^{M\times N}$. We define the effective channel $\mathbf{h}_{\mathrm{eff},(l,k)} \triangleq \mathbf{h}_{\mathrm{D},(l,k)} + \sum_{r\in\mathcal{R}} \mathbf{h}_{(l,r,k)}$. By isolating the term corresponding to the APs serving the k-th UE (i.e., the set $\mathcal{L}_k$), the received signal at the k-th UE can be reformulated as

$$y_k = \sum_{l\in\mathcal{L}_k} \mathbf{h}_{\mathrm{eff},(l,k)} \sum_{i\in\mathcal{K}_l} \mathbf{f}_{(l,i)} s_i + \sum_{l\in\mathcal{L}_k^c} \mathbf{h}_{\mathrm{eff},(l,k)} \sum_{i\in\mathcal{K}_l} \mathbf{f}_{(l,i)} s_i + z_k, \tag{5}$$

where $\mathcal{L}_k^c = \mathcal{L}\setminus\mathcal{L}_k$. By using the fact that $\mathcal{K}_l$ and $\mathcal{K}_l\setminus\{k\}$ are equivalent in the second term of (5) since the k-th UE is not served by any AP in the set $\mathcal{L}_k^c$, and isolating the transmit signals intended for the k-th UE in the first term of (5) and merging the remaining signals into the second term of (5), we have

$$y_k = \underbrace{\sum_{l\in\mathcal{L}_k} \mathbf{h}_{\mathrm{eff},(l,k)}\mathbf{f}_{(l,k)} s_k}_{\text{Desired signal}} + \underbrace{\sum_{l\in\mathcal{L}} \mathbf{h}_{\mathrm{eff},(l,k)} \sum_{i\in\mathcal{K}_l\setminus\{k\}} \mathbf{f}_{(l,i)} s_i}_{\text{Interference}} + \underbrace{z_k}_{\text{Noise}}. \tag{6}$$

The achievable rate for the k-th UE can be expressed as (with derivations omitted for brevity):

$$r_k = \log_2\left(1 + \frac{\left|\sum_{l\in\mathcal{L}_k} \mathbf{h}_{\mathrm{eff},(l,k)}\mathbf{f}_{(l,k)}\right|^2}{\sum_{i\in\mathcal{K}\setminus\{k\}} \left|\sum_{l\in\mathcal{L}_i} \mathbf{h}_{\mathrm{eff},(l,k)}\mathbf{f}_{(l,i)}\right|^2 + \sigma_k^2}\right). \tag{7}$$

Our design objective is to maximize the sum rate of all UEs in the network by jointly optimizing AP beamforming vectors and RIS phase shifts. The problem is formulated as

$$\max_{\mathbf{F},\boldsymbol{\theta}} \quad \sum_{k\in\mathcal{K}} r_k \tag{8a}$$

$$\text{s.t.} \quad \mathbb{E}[\|\mathbf{x}_l\|^2] = \mathrm{tr}(\mathbf{F}_l\mathbf{F}_l^{\mathrm{H}}) \le P_{\max,l}, \quad \forall l\in\mathcal{L}, \tag{8b}$$

$$\boldsymbol{\theta}_r \in \boldsymbol{\Theta}^N, \quad \forall r\in\mathcal{R}, \tag{8c}$$

where the optimization variables are $\boldsymbol{\theta} = \left[\boldsymbol{\theta}_1^{\mathrm{T}},\ldots,\boldsymbol{\theta}_R^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{C}^{RN\times 1}$ and $\mathbf{F} = [\mathbf{F}_1,\ldots,\mathbf{F}_L] \in \mathbb{C}^{M\times\sum_{l\in\mathcal{L}}|\mathcal{K}_l|}$, with $\mathbf{F}_l \in \mathbb{C}^{M\times|\mathcal{K}_l|}$ formed by horizontally concatenating all the vectors in the set $\left\{\mathbf{f}_{(l,k)} \mid k\in\mathcal{K}_l\right\}$. Constraints (8b) and (8c) specify the transmit power budget at the l-th AP and restrict the RIS phase shifts to the set $\boldsymbol{\Theta}$, which can be either a continuous set $\boldsymbol{\Theta}_{\mathrm{cont}} = \left\{e^{j\varphi_n} \mid \varphi_n\in[0,2\pi]\right\}$ or a discrete set $\boldsymbol{\Theta}_{\mathrm{disc}} = \left\{e^{j\varphi_n} \mid \varphi_n \in \left\{0,\frac{2\pi}{2^Q},\ldots,\frac{2\pi(2^Q-1)}{2^Q}\right\}\right\}$ with Q-bit resolution.

Solving Problem (8) in a centralized manner, either by model-based or learning-based methods, requires the collection of CSI between each AP and its served UEs (including all direct channels $\mathbf{h}_{\mathrm{D},(l,k)}$ and reflected channels $\mathbf{h}_{(l,r,k)}$) at the CPU, which then optimizes the AP beamforming vectors and RIS phase configurations. This centralized approach incurs significant signaling overhead due to extensive exchange of CSI, especially in expansive cell-free networks. To mitigate this, we propose a decentralized learning-based solution where each AP independently designs its beamforming vectors and provides RIS phase estimates based solely on locally available CSI. The proposed framework is detailed next.

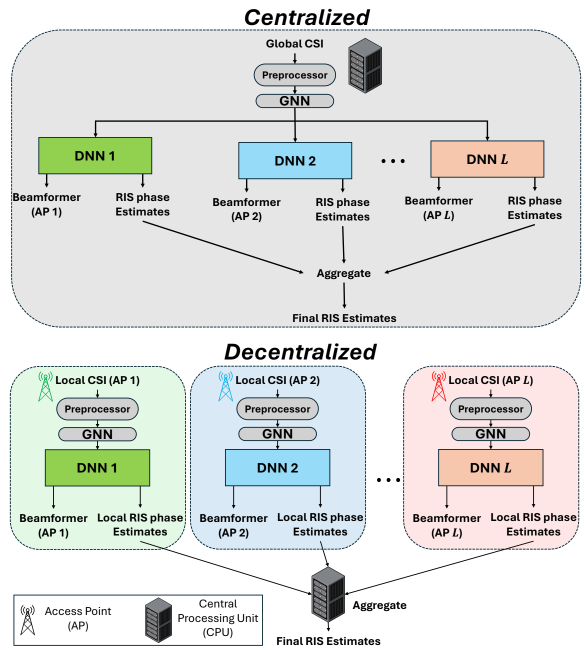

Fig. 2. Centralized vs. decentralized frameworks.

III. Proposed Decentralized Graph Neural Network-Based Method

A. Decentralized Framework

The overall proposed decentralized framework is illustrated in Fig. 2, alongside its centralized counterpart. In the centralized learning approach, a single GNN processes global CSI input, followed by simple DNN readout modules that generate the beamformer for each AP and the corresponding RIS phase estimates. These estimates are then aggregated, accounting for the relative importance of each RIS to each AP, to determine the final RIS configuration. Both GNN and DNN models are trained centrally at the CPU using global CSI.

In decentralized learning, the trained GNN is replicated at each AP and executed locally using local CSI to independently compute beamforming vectors and estimate RIS phase shifts. The per-AP RIS phase estimates are then sent to the CPU to determine the final RIS configuration, since RISs serve the entire network and cannot be configured by a single AP.

B. Local CSI Acquisition

Here, we describe the local CSI acquisition for decentralized learning. First define

$$\tilde{\mathbf{h}}_{(l,r,k)} \triangleq \begin{bmatrix} \mathrm{Re}\{\mathrm{vec}(\widetilde{\mathbf{H}}_{(l,r,k)})\} \\ \mathrm{Im}\{\mathrm{vec}(\widetilde{\mathbf{H}}_{(l,r,k)})\} \end{bmatrix} \in \mathbb{R}^{2M(N+1)\times 1}, \tag{9}$$

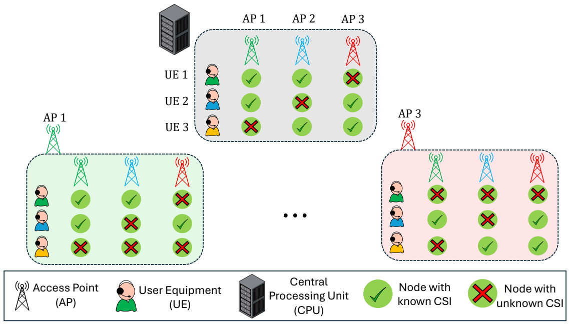

Fig. 3. An example with $L = 3$ APs and $K = 3$ UEs illustrating global CSI at the CPU and local CSI at each AP. The local CSI is a subset of the global CSI and varies across APs.

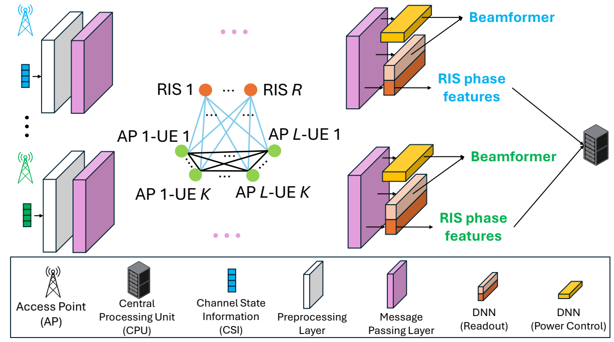

Fig. 4. GNN architecture.

where $\widetilde{\mathbf{H}}_{(l,r,k)} = \left[\mathbf{H}^{\mathrm{T}}_{(l,r,k)}, \mathbf{h}^{\mathrm{T}}_{\mathrm{D},(l,k)}\right] \in \mathbb{C}^{M\times(N+1)}$. It is assumed that each UE can extract $\widetilde{\mathbf{h}}_{(l,r,k)}$ from the effective channel $\mathbf{h}_{\mathrm{eff},(l,k)}$ during the downlink pilot transmission phase with controlled RIS configurations and feed it back to the associated APs. The set of CSI vectors available at the l-th AP is given by

$$\mathcal{H}^{(l)} = \left\{\widetilde{\mathbf{h}}_{(l,r,k)} \mid k \in \mathcal{K}_l, r \in \mathcal{R}\right\} \bigcup \left\{\widetilde{\mathbf{h}}_{(l',r,k)} \mid k \in \mathcal{K}_l \text{ and } l' \in \mathcal{L}_k, r \in \mathcal{R}\right\}. \quad (10)$$

Note that the first set in (10) is obtained via feedback from UEs to the serving l-th AP, while the second set arises from allowing UEs to additionally report CSI of their links with other serving APs (the l'-th AP, $l' \in \mathcal{L}_k$). This is illustrated in the example below.

An Illustrative Example: Consider an example shown in Fig. 3 with three APs and three UEs. We assume that UE 1 is served by AP 1 and AP 2, UE 2 is served by AP 1 and AP 3, and UE 3 is served by AP 2 and AP 3. From the centralized perspective, the CPU has access to all CSI corresponding to the aforementioned AP–UE associations, i.e., global CSI. Now, consider the CSI available locally at AP 1 during the decentralized inference phase. Since AP 1 serves UE 1 and UE 2, the CSI vectors $\widetilde{\mathbf{h}}_{(1,r,1)}$ and $\widetilde{\mathbf{h}}_{(1,r,2)}$ for all $r \in \mathcal{R}$ are estimated by UE 1 and UE 2, respectively, and fed back to AP 1. To further improve the inference capability at AP 1, we exploit the FDD nature of the system and allow UEs served by AP 1 to additionally feed back CSI corresponding to their links with other serving APs. Specifically, since UE 1 is also served by AP 2, the CSI vectors $\widetilde{\mathbf{h}}_{(2,r,1)}$ for all $r \in \mathcal{R}$ can be estimated by UE 1 and subsequently fed back to AP 1. Similarly, since UE 2 is served by both AP 1 and AP 3, the CSI vectors $\widetilde{\mathbf{h}}_{(3,r,2)}$ for all $r \in \mathcal{R}$ is also fed back to AP 1. This yields the total local CSI at AP 1, including $\widetilde{\mathbf{h}}_{(1,r,1)}$ and $\widetilde{\mathbf{h}}_{(1,r,2)}$ (from the first set in (10)), and $\widetilde{\mathbf{h}}_{(2,r,1)}$ and $\widetilde{\mathbf{h}}_{(3,r,2)}$ (from the second set in (10)), as shown in Fig. 3. The same feedback mechanism is applied to all APs to obtain the local CSI for decentralized learning.

C. GNN Architecture

The proposed GNN architecture is illustrated in Fig. 4 and described as follows.

1) Graph Representation: The network is modeled as a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the node and edge sets, respectively. The node set $\mathcal{V}$ consists of $L \times K$ AP–UE pair nodes and R RIS nodes, resulting in a total of $L \times K + R$ nodes. The edge set $\mathcal{E}$ consists of two types of edges: unweighted edges among AP–UE nodes, and weighted edges between RIS nodes and AP–UE nodes, denoted as $e_{(l,r,k)} \triangleq \mathrm{tr}\left(\mathbf{H}^{\mathrm{T}}_{(l,r,k)}\mathbf{H}^{*}_{(l,r,k)}\right)$ for the edge connecting the RIS r node and the AP l–UE k node. The weighted edge $e_{(l,r,k)}$ captures the relative importance of RIS r for the AP l–UE k link, which is encoded in $\mathbf{H}_{(l,r,k)}$.

2) Preprocessing Stage: Since the CSI vectors are processed in the same manner regardless of the AP, we describe the procedure for the l-th AP. As shown in Fig. 2, the CSI is first preprocessed before being fed into the GNN. Each vector $\widetilde{\mathbf{h}}_{(l,r,k)} \in \mathcal{H}^{(l)}$ is independently passed through a neural network $f^{(0)} : \mathbb{R}^{2M(N+1)\times 1} \to \mathbb{R}^{q\times 1}$ as follows:

$$f^{(0)} : \widetilde{\mathbf{h}}_{(l,r,k)} \mapsto \mathbf{W}^{(0)}_2 \phi_{\mathrm{LReLU}}\left(\mathbf{W}^{(0)}_1 \widetilde{\mathbf{h}}_{(l,r,k)}\right), \quad (11)$$

where $\mathbf{W}^{(0)}_1 \in \mathbb{R}^{2q\times 2M(N+1)}$ and $\mathbf{W}^{(0)}_2 \in \mathbb{R}^{q\times 2q}$ are trainable weights, and ϕ_{LReLU} is the element-wise Leaky ReLU function. Based on the outputs of $f^{(0)}$, we obtain the set of initial feature vectors for AP–UE nodes $\mathcal{U}^{(0)} = \left\{\mathbf{u}^{(0)}_{(l,k)} \mid (l,k) \in \mathcal{L} \times \mathcal{K}\right\}$, where $\mathbf{u}^{(0)}_{(l,k)} = \frac{1}{R}\sum_{r\in\mathcal{R}} f^{(0)}(\widetilde{\mathbf{h}}_{(l,r,k)})$.

The initial feature vectors for the RIS nodes are derived as follows. First, define $a_{(l,k)} \triangleq \mathbf{h}_{\mathrm{D},(l,k)}\mathbf{h}^{\mathrm{H}}_{\mathrm{D},(l,k)}$ and

$$\mathbf{p}_r = \begin{bmatrix} \frac{1}{c_{1,r}} \sum_{l\in\mathcal{L},\, k\in\mathcal{K}} e_{(l,r,k)} \mathbf{u}^{(0)}_{(l,k)} \\ \frac{1}{c_2} \sum_{l\in\mathcal{L},\, k\in\mathcal{K}} a_{(l,k)} \mathbf{u}^{(0)}_{(l,k)} \end{bmatrix} \in \mathbb{R}^{2q\times 1}, \quad (12)$$

where $c_{1,r} = \sum_{l\in\mathcal{L},\, k\in\mathcal{K}} e_{(l,r,k)}$ and $c_2 = \sum_{l\in\mathcal{L},\, k\in\mathcal{K}} a_{(l,k)}$ are normalization factors. Here, $\mathbf{p}_r$ represents a weighted sum of initial AP–UE node features $\mathbf{u}^{(0)}_{(l,k)}$, with weights accounting for both direct channels (via $a_{(l,k)}$) and reflected channels (via $e_{(l,r,k)}$). The inclusion of direct channels is motivated by the observation that stronger direct links reduce the dependence on

RIS-assisted communication. Each vector $\mathbf{p}_r$ is independently sent through another neural network $f_{\text{RIS}}^{(0)}\colon \mathbb{R}^{2q\times 1} \rightarrow \mathbb{R}^{q\times 1}$ as

$$f_{\text{RIS}}^{(0)}\colon \mathbf{p}_r \mapsto \mathbf{W}_{\text{RIS},2}^{(0)}\phi_{\text{LReLU}}\big(\mathbf{W}_{\text{RIS},1}^{(0)}\mathbf{p}_r\big), \tag{13}$$

where $\mathbf{W}_{\text{RIS},1}^{(0)} \in \mathbb{R}^{2q\times 2q}$ and $\mathbf{W}_{\text{RIS},2}^{(0)} \in \mathbb{R}^{q\times 2q}$ are trainable weights. The resulting set of initial RIS feature vectors is given by $\mathcal{S}^{(0)} = \left\{\mathbf{s}_r^{(0)} \mid r \in \mathcal{R}\right\}$, where $\mathbf{s}_r^{(0)} = f_{\text{RIS}}^{(0)}(\mathbf{p}_r)$. The sets $\mathcal{U}^{(0)}$ and $\mathcal{S}^{(0)}$ form the inputs to the GNN.

3) Message Passing and Readout Stages: The GNN performs a series of message-passing operations followed by a final readout layer implemented by the DNN. Eventually, the RIS r node learns a local estimate of $\boldsymbol{\theta}_r$ and the AP l–UE k node learns $\mathbf{f}_{(l,k)}$.

Message Passing: At the d-th message-passing layer, where $d = 1, \ldots, D$, the feature vectors of AP–UE nodes are updated through aggregation and combination. With the feature set $\mathcal{U}^{(d-1)}$ from the previous layer, we compute for the AP l–UE k node:

$$\mathbf{u}_{\max,(l,k)}^{(d-1)} = \max_{\forall (i,j)\neq(l,k)} \mathbf{u}_{(i,j)}^{(d-1)}, \tag{14}$$

$$\mathbf{s}_{\text{mean},(l,k)}^{(d-1)} = \frac{1}{c_{3,(l,k)}} \sum_{r\in\mathcal{R}} e_{(l,r,k)}\mathbf{s}_r^{(d-1)}, \tag{15}$$

where the $\max$ operator is applied element-wise, and $c_{3,(l,k)} = \sum_{r\in\mathcal{R}} e_{(l,r,k)}$. The vectors $\mathbf{u}_{(l,k)}^{(d-1)}$, $\mathbf{u}_{\max,(l,k)}^{(d-1)}$, and $\mathbf{s}_{\text{mean},(l,k)}^{(d-1)}$ are then concatenated to form

$$\widetilde{\mathbf{u}}_{(l,k)}^{(d-1)} = \begin{bmatrix} \mathbf{u}_{(l,k)}^{(d-1)} \\ \mathbf{u}_{\max,(l,k)}^{(d-1)} \\ \mathbf{s}_{\text{mean},(l,k)}^{(d-1)} \end{bmatrix} \in \mathbb{R}^{3qd\times 1}. \tag{16}$$

The vector $\widetilde{\mathbf{u}}_{(l,k)}^{(d-1)}$ is then passed through the mapping $f^{(d)}\colon \mathbb{R}^{3qd\times 1} \rightarrow \mathbb{R}^{q\times 1}$ as follows:

$$f^{(d)}\colon \widetilde{\mathbf{u}}_{(l,k)}^{(d-1)} \mapsto \mathbf{W}_2^{(d)}\phi_{\text{LReLU}}\big(\mathbf{W}_1^{(d)}\widetilde{\mathbf{u}}_{(l,k)}^{(d-1)}\big), \tag{17}$$

where $\mathbf{W}_1^{(d)} \in \mathbb{R}^{2q\times 3qd}$ and $\mathbf{W}_2^{(d)} \in \mathbb{R}^{q\times 2q}$ are trainable weights. To preserve the information from the previous layer, the output $f^{(d)}(\widetilde{\mathbf{u}}_{(l,k)}^{(d-1)})$ is concatenated with $\mathbf{u}_{(l,k)}^{(d-1)}$ to form the representation at the d-th layer:

$$\mathbf{u}_{(l,k)}^{(d)} = \begin{bmatrix} f^{(d)}(\widetilde{\mathbf{u}}_{(l,k)}^{(d-1)}) \\ \mathbf{u}_{(l,k)}^{(d-1)} \end{bmatrix} \in \mathbb{R}^{q(d+1)\times 1}. \tag{18}$$

The set of feature vectors for the AP–UE nodes at the d-th layer is represented as $\mathcal{U}^{(d)} = \big\{\mathbf{u}_{(l,k)}^{(d)} \mid (l,k) \in \mathcal{L}\times\mathcal{K}\big\}$.

The feature vectors of RIS nodes are updated similarly. For the RIS r node, we define

$$\widetilde{\mathbf{s}}_r^{(d-1)} = \begin{bmatrix} \mathbf{s}_r^{(d-1)} \\ \frac{1}{c_{1,r}} \sum_{l\in\mathcal{L},k\in\mathcal{K}} e_{(l,r,k)}\mathbf{u}_{(l,k)}^{(d-1)} \\ \frac{1}{c_2} \sum_{l\in\mathcal{L},k\in\mathcal{K}} a_{(l,k)}\mathbf{u}_{(l,k)}^{(d-1)} \end{bmatrix} \in \mathbb{R}^{3qd\times 1}. \tag{19}$$

The vector $\widetilde{\mathbf{s}}_r^{(d-1)}$ is then passed through the mapping $f_{\text{RIS}}^{(d)}\colon \mathbb{R}^{3qd\times 1} \rightarrow \mathbb{R}^{q\times 1}$ defined as

$$f_{\text{RIS}}^{(d)}\colon \widetilde{\mathbf{s}}_r^{(d-1)} \mapsto \mathbf{W}_{\text{RIS},2}^{(d)}\phi_{\text{LReLU}}\big(\mathbf{W}_{\text{RIS},1}^{(d)}\widetilde{\mathbf{s}}_r^{(d-1)}\big), \tag{20}$$

where $\mathbf{W}_{\text{RIS},1}^{(d)} \in \mathbb{R}^{2q\times 3qd}$ and $\mathbf{W}_{\text{RIS},2}^{(d)} \in \mathbb{R}^{q\times 2q}$ are trainable weights. The representation of the RIS r node at the d-th layer is then given by

$$\mathbf{s}_r^{(d)} = \begin{bmatrix} f_{\text{RIS}}^{(d)}(\widetilde{\mathbf{s}}_r^{(d-1)}) \\ \mathbf{s}_r^{(d-1)} \end{bmatrix} \in \mathbb{R}^{q(d+1)\times 1}. \tag{21}$$

Finally, the set of feature vectors for the RIS nodes at the d-th layer is $\mathcal{S}^{(d)} = \big\{\mathbf{s}_r^{(d)} \mid r \in \mathcal{R}\big\}$.

Readout: A readout layer is applied after the D message-passing layers producing $\mathcal{U}^{(D)}$ and $\mathcal{S}^{(D)}$. The readout layer is implemented by a DNN at each AP to generate the AP beamforming vector $\mathbf{f}_{(l,k)}$ and per-AP estimate of $\boldsymbol{\theta}_r$. To obtain the beamforming vector, each vector in $\mathcal{U}^{(D)}$ is passed through an AP readout layer $f^{\text{out}}\colon \mathbb{R}^{q(D+1)\times 1} \rightarrow \mathbb{R}^{2M\times 1}$, defined as

$$f^{\text{out}}\colon \mathbf{u}_{(l,k)}^{(D)} \mapsto \mathbf{W}^{\text{out}}\mathbf{u}_{(l,k)}^{(D)}, \tag{22}$$

where $\mathbf{W}^{\text{out}} \in \mathbb{R}^{2M\times q(D+1)}$ denotes the trainable weight matrix. To ensure the beamforming output meets constraint (8b), normalization constant α_l is applied for each AP l:

$$\alpha_l = \phi_{\text{sig}}\Bigg(\frac{1}{LK}\mathbf{W}_{\text{pow}}\Big(\sum_{(l,k)\in\mathcal{L}\times\mathcal{K}} \xi\big(\mathbf{u}_{(l,k)}^{(D)}\big)\Big)\Bigg) \in [0,1], \tag{23}$$

where $\mathbf{W}_{\text{pow}} \in \mathbb{R}^{1\times q}$ is the trainable weight matrix, $\phi_{\text{sig}}\colon \mathbb{R} \rightarrow \mathbb{R}$ is the sigmoid activation function, and $\xi\colon \mathbb{R}^{q(D+1)\times 1} \rightarrow \mathbb{R}^{q\times 1}$ is specified as

$$\xi\colon \mathbf{u}^{(D)} \mapsto \mathbf{W}_{\xi,2}\phi_{\text{LReLU}}\big(\mathbf{W}_{\xi,1}\mathbf{u}^{(D)}\big), \tag{24}$$

with $\mathbf{W}_{\xi,1} \in \mathbb{R}^{2q\times q(D+1)}$ and $\mathbf{W}_{\xi,2} \in \mathbb{R}^{q\times 2q}$ being trainable weight matrices. The beamforming vector $\mathbf{f}_{(l,k)}$ is finally obtained at the readout layer as

$$\begin{bmatrix} \text{Re}\{\mathbf{f}_{(l,k)}\} \\ \text{Im}\{\mathbf{f}_{(l,k)}\} \end{bmatrix} = \begin{cases} \sqrt{\frac{P_{\max,l}\alpha_l}{\sum_{k\in\mathcal{K}_l}\left\|f^{\text{out}}(\mathbf{u}_{(l,k)}^{(D)})\right\|_2^2}} f^{\text{out}}\big(\mathbf{u}_{(l,k)}^{(D)}\big), & \text{if } k \in \mathcal{K}_l \\ \mathbf{0}, & \text{otherwise} \end{cases}. \tag{25}$$

To obtain the per-AP estimate of RIS phases, each AP employs its own RIS readout layer $f_{\text{RIS},l}^{\text{out}}\colon \mathbb{R}^{q(D+1)\times 1} \times \mathbb{R}^{RK\times 1} \rightarrow \mathbb{R}^{4N\times 1}$, defined as

$$f_{\text{RIS},l}^{\text{out}}\colon (\mathbf{s}^{(D)}, \mathbf{e}_l) \mapsto \begin{bmatrix} \mathbf{W}_{\text{RIS},1}^{\text{out}}\mathbf{s}_r^{(D)} \\ \mathbf{W}_{\text{RIS},2}^{\text{out}}\mathbf{e}_l \end{bmatrix}, \tag{26}$$

where $\mathbf{e}_l$ is a vector of size $RK \times 1$ with entries from the set $\{e_{(l,r,k)} \mid r \in \mathcal{R}, k \in \mathcal{K}\}$, and $\mathbf{W}_{\text{RIS},1}^{\text{out}} \in \mathbb{R}^{2N\times q(D+1)}$ and $\mathbf{W}_{\text{RIS},2}^{\text{out}} \in \mathbb{R}^{2N\times RK}$ are trainable weight matrices. Each vector in $\mathcal{S}^{(D)}$, along with $\mathbf{e}_l$, is then passed through $f_{\text{RIS},l}^{\text{out}}$ to produce the set $\{f_{\text{RIS},l}^{\text{out}}(\mathbf{s}_r^{(D)}, \mathbf{e}_l) \mid r \in \mathcal{R}\}$. Finally, to obtain the final RIS phase shifts, the per-AP estimates at all APs $\{f_{\text{RIS},l}^{\text{out}}(\mathbf{s}_r^{(D)}, \mathbf{e}_l) \mid l \in \mathcal{L}\}$ are collected in $\mathbf{R}_r = \big[f_{\text{RIS},1}^{\text{out}}(\mathbf{s}_r^{(D)}, \mathbf{e}_1), \ldots, f_{\text{RIS},L}^{\text{out}}(\mathbf{s}_r^{(D)}, \mathbf{e}_L)\big] \in \mathbb{R}^{4N\times L}$ and aggregated at the CPU via the function $f_{\text{RIS}}^{\text{aggre}}: \mathbb{R}^{4N\times L} \rightarrow \mathbb{R}^{2N\times 1}$, which combines the RIS phase features from all APs to determine the phase shifts for the r-th RIS, $\boldsymbol{\theta}_r$:

$$f_{\text{RIS}}^{\text{aggre}}: \mathbf{R}_r \mapsto f_{\text{norm}}(\mathbf{W}^{\text{reduce}}\mathbf{R}_r\mathbf{1}_L), \tag{27}$$

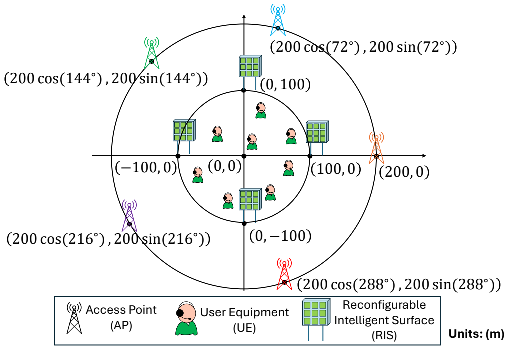

Fig. 5. Experimental setup.

where $\mathbf{1}_L$ denotes an all-ones vector of dimension L, used to aggregate the columns of $\mathbf{R}_r$, $\mathbf{W}^{\text{reduce}} \in \mathbb{R}^{2N\times 4N}$ is a trainable weight matrix, and $f_{\text{norm}} : \mathbb{R}^{2N\times 1} \to \mathbb{R}^{2N\times 1}$ performs normalization such that constraint (8c) is satisfied. The final RIS phase shifts $\boldsymbol{\theta}_r$ for the continuous-phase case are obtained as

$$\begin{bmatrix} \mathrm{Re}\{\boldsymbol{\theta}_r\} \\ \mathrm{Im}\{\boldsymbol{\theta}_r\} \end{bmatrix} = f_{\text{RIS}}^{\text{aggre}}(\mathbf{R}_r). \tag{28}$$

For the discrete-phase case, each $\theta_{(r,n)}$ in $\boldsymbol{\theta}_r$ is further quantized to the closest point in $\boldsymbol{\Theta}_{\text{disc}}$.

IV. Simulation Results and Analysis

We consider the setup in Fig. 5, where $L = 5$ APs are placed evenly on a circle of radius 200 m, $R = 4$ RISs on a circle of radius 100 m, and $K = 8$ UEs are randomly distributed within the circle of radius 100 m. The l-th AP serves the k-th UE if $\left\|\mathbf{h}_{\mathrm{D},(l,k)}^{\mathrm{T}}\right\|_2^2 \geq \rho \max_{l'\in\mathcal{L}} \left\|\mathbf{h}_{\mathrm{D},(l',k)}^{\mathrm{T}}\right\|_2^2$, where $\rho \in [0, 1]$ is the AP–UE association threshold. The default simulation parameters are listed in Table I. The networks are trained for 2000 iterations using Adam optimizer with learning rate 10^{-4}, weight decay 10^{-6}, and batch size 8. We compare the proposed decentralized method with the centralized scheme under continuous ("C"), discrete ("D"), and random discrete ("R-D") RIS phase settings. For the "R-D" setting, both beamforming vectors and RIS phases are treated as optimization variables during training, while at inference the RIS phases are replaced by random selections from $\boldsymbol{\Theta}_{\text{disc}}$.

Fig. 6 shows the sum rate vs. the number of AP antennas M. All schemes exhibit monotonic performance gains with increasing M. The Centralized (C) scheme achieves the highest performance due to access to global CSI, while the Decentralized (C) scheme performs closely, demonstrating the effectiveness of the proposed decentralized approach. The small performance gap indicates that, with modest additional CSI feedback in the proposed local CSI acquisition strategy, near-centralized performance can be achieved with significantly reduced backhaul overhead. Discrete RIS phases incur slight performance loss due to limited resolution, whereas random RIS phases perform substantially worse, highlighting the importance of optimizing RIS-assisted links and the effectiveness of the proposed aggregation scheme to determine the final RIS phase configurations.

TABLE I
Simulation Parameters

Parameter	Symbol	Value
Number of APs	L	5
Number of AP antennas	M	2
Number of RISs	R	4
Number of RIS elements	N	30
Number of UEs	K	8
Rician factors	$\kappa_{\mathrm{AR},(l,r)}, \kappa_{\mathrm{RU},(r,k)}$	10
Large-scale fading gains (AP–RIS)	$\bar{\eta}_{\mathrm{AR},(l,r)}, \bar{\bar{\eta}}_{\mathrm{AR},(l,r)}$	$10^{-0.5}$
LoS, NLoS path-loss exp. (AP–RIS)	$\bar{\gamma}_{\mathrm{AR},(l,r)}, \bar{\bar{\gamma}}_{\mathrm{AR},(l,r)}$	$2, 2.5$
Large-scale fading gains (RIS–UE)	$\bar{\eta}_{\mathrm{RU},(r,k)}, \bar{\bar{\eta}}_{\mathrm{RU},(r,k)}$	$10^{-0.5}$
LoS, NLoS path loss exp. (RIS–UE)	$\bar{\gamma}_{\mathrm{RU},(r,k)}, \bar{\bar{\gamma}}_{\mathrm{RU},(r,k)}$	$2, 2.5$
Large-scale fading gains (AP–UE)	$\bar{\bar{\eta}}_{\mathrm{D},(r,k)}$	$10^{-4.5}$
NLoS path loss exp. (AP–UE)	$\bar{\bar{\gamma}}_{\mathrm{D},(l,k)}$	3.5
AP–UE association threshold	ρ	0.1
Discrete RIS phase resolution	Q	2
AP transmit power budget	$P_{\max,l}$	15 dBm
Number of GNN layers	D	6
GNN latent parameter	q	64

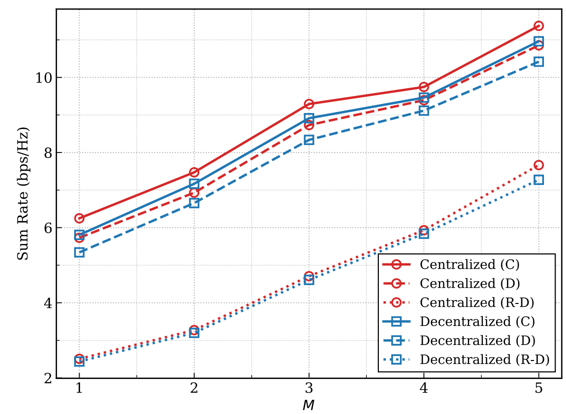

Fig. 6. Sum-rate vs. the number of AP antennas M.

Fig. 7 shows the sum rate vs. the AP transmit power budget $P_{\max,l} = P_{\max}, \forall l$. All schemes show increased performance with increasing $P_{\max}$, while discrete and random discrete RIS phases exhibit diminishing gains at higher $P_{\max}$, emphasizing the importance of RIS configuration at higher power levels. The decentralized schemes closely match their centralized counterparts, nearly overlapping at higher $P_{\max}$, which demonstrates the effectiveness of the proposed decentralized approach.

Table II compares the signaling overhead of decentralized and centralized approaches in terms of the number of real coefficients exchanged between the APs and the CPU. First, consider the signaling overhead from the APs to the CPU. For the centralized scheme, this consists of all CSI estimated at the UEs and forwarded to the APs and then the CPU. The CSI associated with the k-th UE, $\mathbf{h}_{\text{eff},(l,k)} \in \mathbb{C}^{1\times M}$ for $l \in \mathcal{L}_k$, contains $2M|\mathcal{L}_k|$ real coefficients. Summing over all UEs, the total overhead is $2M\sum_{k=1}^{K}|\mathcal{L}_k|$ real coefficients. For the decentralized scheme, no CSI is sent to the CPU. The AP-to-CPU overhead arises from RIS phase features transmitted for aggregation. The features for the r-th RIS from all L APs, $\mathbf{R}_r \in \mathbb{R}^{4N\times L}$, contain $4NL$ real coefficients, yielding a total

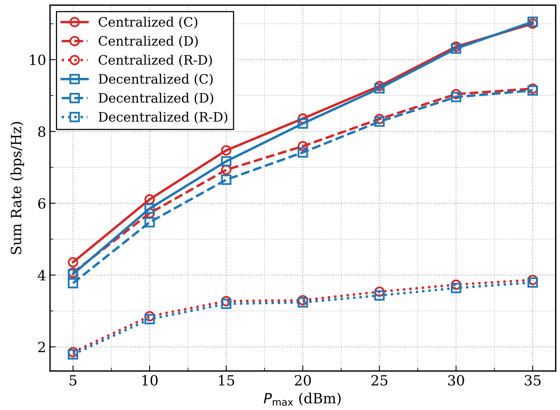

Fig. 7. Sum-rate vs. the AP transmit power budget $P_{\max,l} = P_{\max}, \forall l$.

TABLE II
SIGNALING OVERHEAD COMPARISON (UNIT: NUMBER OF REAL COEFFICIENTS)

	Centralized	**Decentralized**
APs to CPU	$2M\sum_{k=1}^{K}\lvert\mathcal{L}_k\rvert$	$4RNL$
CPU to APs and RISs	$2M\sum_{l=1}^{L}\lvert\mathcal{K}_l\rvert + 2RN$	$2RN$

overhead of $4RNL$ real coefficients across all RISs.

Next, consider the signaling overhead from the CPU to the APs and RISs. For the centralized scheme, the optimized beamforming matrices are sent from the CPU to the APs. The beamforming matrix for the l-th AP, $\mathbf{F}_l \in \mathbb{C}^{M\times|\mathcal{K}_l|}$, contains $2M\,|\mathcal{K}_l|$ real coefficients, yielding a total of $2M\sum_{l=1}^{L}|\mathcal{K}_l|$ for all APs. In contrast, the decentralized scheme incurs no such signaling exchange, as beamforming is computed locally at each AP. For both schemes, RIS configurations are set by the CPU, resulting in the same signaling overhead of $2RN$ real coefficients for $\boldsymbol{\theta} \in \mathbb{C}^{RN\times 1}$.

Overall, the advantage of the decentralized scheme is that the AP-to-CPU overhead does not scale with the number of UEs, making it well-suited for large cell-free networks, and the CPU-to-AP overhead which scales with the number of APs is completely avoided. However, it introduces additional AP-to-CPU overhead for RIS configuration, since RISs serve the entire network and cannot be managed by individual APs, necessitating an extra coordination mechanism in the decentralized framework. Mitigating this overhead is a worthwhile direction for future work.

V. CONCLUSION

In this work, we proposed a decentralized GNN-based framework for multi-RIS-aided multiuser cell-free networks. The proposed method decouples the learning tasks across APs using only locally available CSI to independently compute AP beamforming and RIS phase estimates. Simulation results and overhead analysis showed that the decentralized approach achieves sum-rate performance comparable to its centralized counterpart, while reducing AP-to-CPU and CPU-to-AP/RIS signaling overhead. Future work includes designing a decoupled learning mechanism for RIS configuration to further reduce aggregation overhead.

REFERENCES

[1] H. Q. Ngo, A. Ashikhmin, H. Yang, E. G. Larsson, and T. L. Marzetta, "Cell-free massive MIMO versus small cells," *IEEE Trans. Wireless Commun.*, vol. 16, no. 3, pp. 1834–1850, Mar. 2017.

[2] E. Björnson and L. Sanguinetti, "Scalable cell-free massive MIMO systems," *IEEE Trans. Commun.*, vol. 68, no. 7, pp. 4247–4261, Jul. 2020.

[3] T. Wang, J. Li, K. Xie, L. Gao, Y. Jiang, and J. Luo, "Low-complexity joint beamforming design for intelligent reflecting surface aided cell-free massive MIMO systems," *IEEE Trans. Veh. Technol.*, pp. 1–15, Oct. 2025 (early access).

[4] Y. Zhang, B. Di, H. Zhang, J. Lin, Y. Li, and L. Song, "Reconfigurable intelligent surface aided cell-free MIMO communications," *IEEE Wireless Commun. Lett.*, vol. 10, no. 4, pp. 775–779, Apr. 2021.

[5] G. Zhang, Y. Lu, L. Zhu, W. Chen, Z. Zhong, and T. Q. S. Quek, "Hybrid beamforming design for RIS-aided full-duplex cell-free networks," *IEEE Trans. Commun.*, vol. 73, no. 9, pp. 7985–8000, Sep. 2025.

[6] C. Xu, Y. Jia, Y. Chen, and W. Huang, "Joint AP clustering and beamforming design for RIS-aided cell-free networks," *IEEE Trans. Veh. Technol.*, vol. 74, no. 5, pp. 8315–8320, May 2025.

[7] H. Ke, J. Xu, W. Xu, C. Yuen, and Z. Lu, "Joint RIS-UE association and beamforming design in RIS-assisted cell-free MIMO network," *IEEE Trans. Commun.*, vol. 73, no. 11, pp. 10 616–10 630, Nov. 2025.

[8] Y. Wang, J. Jiang, X. Du, Z. Lian, Q. Wu, and W. Chen, "Efficient joint precoding design for wideband intelligent reflecting surface-assisted cell-free network," *IEEE Trans. Commun.*, vol. 73, no. 8, pp. 6914–6927, Aug. 2025.

[9] Z. Gao, R. Zhong, X. Mu, Z. Du, J. Liu, and Y. Liu, "Joint beamforming design in multi-STAR-RISs-aided cell-free massive MIMO networks," *IEEE Internet Things J.*, vol. 13, no. 7, pp. 13 965–13 981, Apr. 2026.

[10] Y. Lu, J. Zhang, Y. Zhu, J. Zheng, D. Yang, and D. W. K. Ng, "Performance optimization of RIS-aided cell-free massive MIMO systems with DRL approach," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 6711–6728, 2026.

[11] T. D. Hua, M. Mohammadi, H. Q. Ngo, and M. Matthaiou, "Cell-free massive MIMO SWIPT with beyond diagonal reconfigurable intelligent surfaces," *IEEE Trans. Commun.*, vol. 73, no. 12, pp. 13 340–13 356, Dec. 2025.

[12] Q.-S. Nguyen, X.-T. Dang, and O.-S. Shin, "Optimization of RIS-assisted cell-free massive MIMO systems with heterogeneous graph neural networks under imperfect channel estimation," *IEEE Trans. Veh. Technol.*, pp. 1–16, Oct. 2025 (early access).

[13] J. Du, S. Gong, H. Liu, T. Yu, F. Jiang, and C. Xing, "Robust beamforming design for active sub-connected RIS assisted cell-free MIMO systems: A two-stage distributed approach," *IEEE Internet Things J.*, vol. 12, no. 15, pp. 31 333–31 348, Aug. 2025.

[14] S. Huang, Y. Ye, M. Xiao, H. V. Poor, and M. Skoglund, "Decentralized beamforming design for intelligent reflecting surface-enhanced cell-free networks," *IEEE Wireless Commun. Lett.*, vol. 10, no. 3, pp. 673–677, Mar. 2021.

[15] H. Hojatian, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau, "Decentralized beamforming for cell-free massive MIMO with unsupervised learning," *IEEE Commun. Lett.*, vol. 26, no. 5, pp. 1042–1046, May 2022.

[16] M. Liu, C. Huang, A. Alhammadi, M. Di Renzo, M. Debbah, and C. Yuen, "Beamforming design and association scheme for multi-RIS multi-user mmWave systems through graph neural networks," *IEEE Trans. Wireless Commun.*, vol. 24, no. 9, pp. 7940–7954, Sep. 2025.

[17] K.-L. Chan, R. Y. Chang, F.-T. Chien, and H. V. Poor, "Beamforming and load-balanced user association in RIS-aided mmWave systems via adaptive attention graph neural networks," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 5781–5796, 2026.

[18] T. Jiang, H. V. Cheng, and W. Yu, "Learning to reflect and to beamform for intelligent reflecting surface with implicit channel estimation," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 7, pp. 1931–1945, Jul. 2021.

[19] T.-J. Yeh, W.-C. Tsai, C.-W. Chen, and A.-Y. Wu, "Enhanced-GNN with angular CSI for beamforming design in IRS-assisted mmWave communication systems," *IEEE Commun. Lett.*, vol. 28, no. 4, pp. 827–831, Apr. 2024.